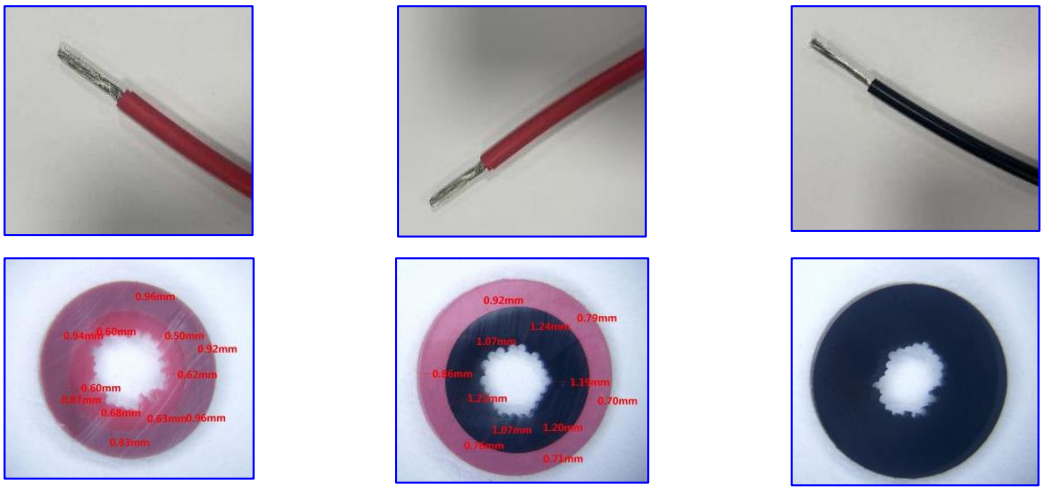
☐ CTL Provisional DECISION SHEET (PDSH)
☑ CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	**Subclause(s)**	**Tracking No.**
IEC 62930:2017	7.1 Table 3 Annex B Table B.1	DSH 2234
Category		
CABL		
Subject	**Keywords**	**Developed by**
Special requirements for the cable with non-separable insulation and sheath	- The cable with non-separable insulation and sheath - Test items - Test specimens	ETF 6

Question

It is stated in Note e of Table B.1 in IEC 62930, that "If the insulation and sheath stick together and it is not possible to prepare separated specimens for insulation and sheath according to IEC 60811-501, the tubular test piece shall be tested and the results applied to both insulation and/or sheath as required."



(√) acceptable
Sample a

(√) acceptable
Sample b

(×) unacceptable
Sample c

Fig. 1 Typical images of cross section and sample pieces of PV cable with non-separable insulation and sheath

However, in case of the insulation and sheath stick together and cannot be separated, as illustrated in Fig. 1, some special requirements for the test items in Table 3 and Table B.1 in IEC 62930 also need to be taken into consideration, such as:

1) Measurement of insulation resistance (Table 3-1.4);
2) Long term resistance of insulation to DC (Table 3-1.5);
3) Insulation thickness (Table 3-2.2.1);

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4) Sheath thickness (Table 3-2.2.2);

5) Weathering/UV resistance on sheath (Table 3-8);

6) Damp heat test (Table 3-10);

7) Bending at low temperature for insulation (Table B.1-1.5)

Thus, how to handle these test items for the cable with non-separable insulation and sheath?

Decision

In case of the insulation and sheath stick together and it is not possible to prepare separated specimens for insulation and sheath according to IEC 60811-501, the special requirements for the test items in Table 3 per IEC 62930:2017 as below,

1) Measurement of insulation resistance (Table 3-1.4);
   Requirement: the completed cable shall be tested.
2) Long term resistance of insulation to DC (Table 3-1.5);
   Requirement: the completed cable shall be tested.
3) Insulation thickness (Table 3-2.2.1);
   Requirement:
   a) treat the insulation and sheath as a whole and prepare the thin slice with insulation and sheath as illustrated in Fig. 1.
   b) a measuring microscope is preferred.
   c) to identify the insulation and sheath so as to determine the thickness, the sheath shall be of a colour contrasting with that of the insulation. Therefore, sample a and b in the Fig.1 are acceptable, while sample c is unacceptable, accordingly.
4) Sheath thickness (Table 3-2.2.2);
   Requirement: refer to the test of Insulation thickness.
5) Weathering/UV resistance on sheath (Table 3-8);
   Requirement: refers to Note e of Table B.1.
6) Damp heat test (Table 3-10);
   Requirement: refers to Note e of Table B.1.
7) Bending at low temperature for insulation (Table B.1-1.5)
   Requirement: N/A (not applicable)

Explanatory notes

Enquiry:
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DSH
Publication date: 2024-09-16

Rationale:

1) As mentioned in cl.2.4 of IEC 60227-2: 1997 (cited in IEC 62930:2017, replaced by 63294: 2021),, “Insulation resistance test applies to all cables. It shall be made on the core samples, 5 m in length, previously submitted to the test described in 2.3 or, if this is not applicable, to the test described in 2.2.”.
   The cl. 2.2 in IEC 60227-2: 1997 is “Voltage test carried out on completed cables”, in other words, the completed cable is allowed be used during the test of Insulation resistance. This situation can also apply to the Long term resistance of insulation to DC test.
2) According to Cl. 5.2.1 and Cl. 5.3.1 in IEC62930: 2017, both of insulation material and sheath material are kind of cross-linked compound, and have very similar mechanical characteristics as specified in Table B.1 in Annex B.
3) During the Weathering/UV resistance(Table 3-8) on sheath and Damp heat test(Table 3-10),the test pieces need to be prepared according to IEC 60811-501.
4) As stated in IEC 60811-501, “Tubular test pieces shall be used only when the dimensions of the core are such that it is not possible to prepare dumb-bell test pieces. Dumb-bell test pieces shall be used whenever possible”, It’s recommended to prepare dumb-bell test pieces for cables having core of nominal cross sectional area greater than 16 mm^2.

The definition of terms

Non-separable insulation and sheath , indicates the insulation and sheath stick together and it is not possible to prepare separated specimens for insulation and sheath according to IEC 60811-501.

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